

23CSE335

Block Chain and its Applications

PROJECT REVIEW 1

**Rich Products: Frozen Food Supply Chain
Traceability**

By

CB.SC.U4CSE23613-D.Viritha

CB.SC.U4CSE23259-Darshan R

Faculty

Dr. J.Uma

Assistant Professor (Sr.Gd.)

School of Computing

PROJECT OBJECTIVE

- Provide farm-to-fork traceability for fully cooked shrimp supply chain.
- Ensure lot-level visibility from fishing boats → vendors → Rich's manufacturing → warehouses → retail → consumer.
- Prevent food fraud , counterfeit ingredients and ensure sustainability certification.
- Enable real-time hold, release, and recall confirmation.
- Give consumers the ability to scan QR / enter lot code and trace shrimp origin and journey.
- Enable analytics & forecasting for smarter supply chain

Innovation

- Blockchain + DNA Tagging – Edible DNA markers + blockchain records → authenticate shrimp, prevent fraud, consumer sees lab-verified origin.
- Automated Recall (Smart Contracts) – Auto-trigger recalls on contamination/temperature breach → instant alerts to vendors & consumers.
- Zero-Knowledge Proofs (ZKP) – Prove certifications & authenticity without exposing sensitive supplier data.
- Geo-Tagged DNA – DNA + location metadata → ensures true harvest origin, prevents false region claims.

Modules and its Job

1. Identity & Onboarding Module

- Registers fishing boats, vendors, warehouses, logistics partners.
- Issues unique digital IDs & certificates for secure participation.

2. Transaction Capture & Validation Module

- Records lot details (ID, product code, shipper/receiver IDs, date, temperature).
- Validates with DNA & geo-tag scans before entry into blockchain.

3. Ledger & Smart Contract Module

- Maintains immutable blockchain record of each supply chain stage.
- Smart contracts enable automated hold, release, and recall processes.

4. Consumer & Enterprise UI Module

- Enterprise Dashboard → lot tracing, QA monitoring, recall execution.
- Consumer Portal (QR/lot code) → shrimp origin, authenticity, recall info.

5. Privacy & Certification Module

- Uses Zero-Knowledge Proofs (ZKP) to prove certifications & sustainability without exposing confidential data.

6. Authenticity Module (DNA + Geo-Tagging)

- Attaches edible DNA molecular tags + harvest geo-location metadata.
- Ensures authenticity and prevents ingredient substitution or false origin claims.

7. Analytics & Forecasting Module

- Analyzes blockchain + supply chain data for demand prediction & waste reduction.
- Optimizes cold chain logistics by detecting recurring failures.

TOOLS AND FRAMEWORKS

- Blockchain Platform: Ethereum (private/testnet) – simulates lot-level transactions.
- Smart Contracts: Solidity (Remix IDE) – validates lot codes, temperature, and DNA mock data.
- Frontend: React.js – QR/lot scanner portal for consumers + enterprise dashboard.
- Backend: Node.js/Express – connects frontend with blockchain and handles lot data.
- DNA Tagging: Simulated digital DNA codes – mock verification instead of real edible markers.
- Off-chain Storage: MongoDB / JSON files – store DNA test reports, sustainability certificates.
- Deployment: Local Environment

Thank you

PRESENTED BY

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